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foundation

GreenFuel-ML

Machine Learning for Sustainable Fuel



Learning Objectives

- *Apply machine learning to predict fuel consumption* and recommend changes that reduce fuel use and CO₂ emissions.
- *Build an end-to-end ML pipeline:* collect and clean data, engineer features, train models, and deploy results.
- *Evaluate and compare models* (RMSE, R²) to choose reliable, interpretable solutions for decision making.
- *Translate model outputs into practical insights* by identifying the factors that affect efficiency the most.
- *Create a simple web demo* so non-technical users can test scenarios and see the impact on fuel and emissions.
- *Publish reproducible code and documentation* for transparency and future improvements.



Tools and Technology used

- **Python & Jupyter:** Main coding environment for data analysis and model development.
- **ML Libraries:** scikit-learn, XGBoost, Pandas, NumPy for data processing and algorithms (e.g: regression, tree models).
- **Visualization:** Matplotlib/Seaborn for charts; Streamlit (or similar) for the interactive app interface.
- **Version Control:** Git and GitHub (GitHub repo [Yuvibarot/GreenFuel-ML](#)) to manage code and collaborate.
- **Data Storage:** CSV/Excel or cloud feature store for datasets (as used in typical ML pipelines).

Methodology

- **Data Gathering:** Compile relevant data (e.g., vehicle specs, fuel types, consumption rates) from public sources or simulated logs.
- **Data Cleaning & Prep:** Handle missing values, outliers, and encode categorical features; scale numerical data.
- **Model Training:** Train multiple ML models (linear regression, decision trees, ensemble methods) to predict fuel efficiency from input features.
- **Evaluation:** Use performance metrics (e.g., RMSE, R^2) to compare models.
- **Deployment:** Build a user-friendly app (e.g., with Streamlit) that takes inputs and shows predicted fuel efficiency

Problem Statement:

- **Fuel inefficiency issue:** Conventional fuel use is often suboptimal, leading to higher costs and CO₂ emissions. Improving efficiency is essential for sustainability.
- **Environmental impact:** Transportation fuels account for a large share of greenhouse gases. Predictive models can guide decisions that reduce emissions and save fuel.
- **Industry need:** Companies like Shell seek innovative solutions to meet energy demands while reducing environmental footprint. A data-driven tool can inform greener fuel strategies.

Solution:

- **GreenFuel-ML model:** A supervised learning model predicts fuel efficiency or consumption (e.g., liters per 100 km) given input conditions (engine size, fuel type, etc.).
- **Insights for action:** The tool can highlight which factors (e.g., vehicle grade, fuel blend) most influence efficiency, helping engineers make informed choices.
- **User interface:** Results are accessible through an interactive app, making it easy for non-technical users to input data and view predictions (similar to the Streamlit deployment shown in other ML projects).
- **Open resources:** The complete solution (data pipeline and model) is documented in the GitHub repo and project website for others to review and build upon.

Screenshot of Output:

 **Hydrogen Production Predictor**

Configure System Parameters
Adjust values to see hydrogen production predictions

Latitude ° **Longitude** ° **Solar Irradiance** kWh/m²/day

31 -50.3 1.2

31 -50.3 1.2

Wind Speed m/s **PV Power** kW **Wind Power** kW

17.1 691.8 528.6

17.1 691.8 528.6

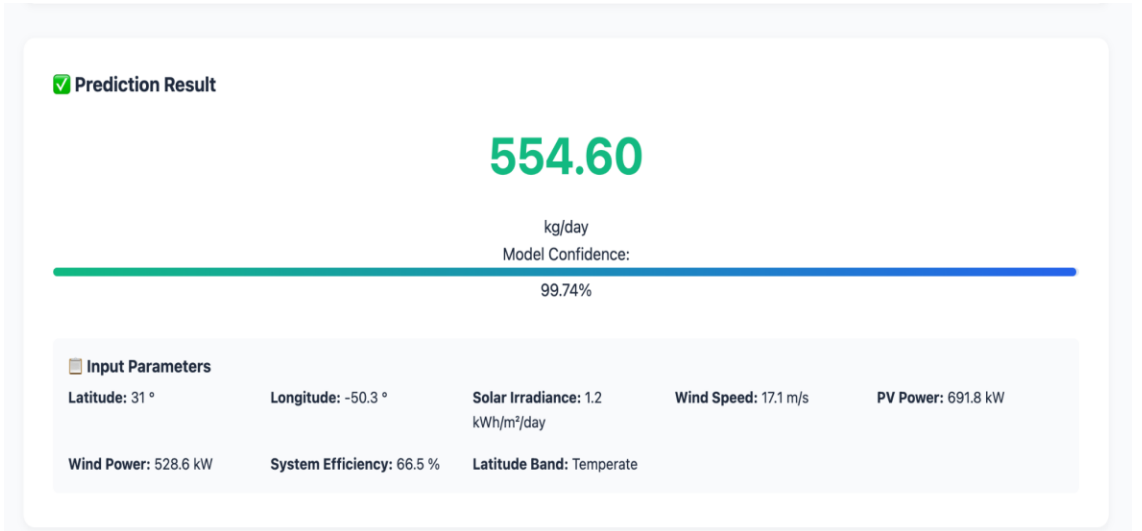
System Efficiency % **Latitude Band**

66.5 Temperate

66.5

Interactive interface allowing users to adjust system parameters such as solar irradiance, wind speed, PV power and system efficiency to see real-time hydrogen production predictions.



Prediction result generated by the GreenFuel-ML model. The system estimates daily hydrogen production based on real-world environmental and system conditions, along with model confidence.

 **Model Information**

 **Performance Metrics**

Metric	Value
R2_SCORE	0.9974
RMSE	8.9965
MAE	5.5288
ACCURACY	96.79

 **Dataset**

Aspect	Value
TOTAL_SAMPLES	2535
TRAINING_SAMPLES	2028
TEST_SAMPLES	507
FEATURES	8

 **Feature Importance**

Model performance metrics. The system demonstrates high predictive reliability with strong accuracy and low error values across the dataset.

Conclusion:

- **Impact:** GreenFuel-ML demonstrates how AI can enhance fuel use efficiency, cutting costs and emissions. This aligns with Shell's mission of providing energy while “minimizing environmental impact”.
- **Educational value:** The project exemplifies Edunet's aim to promote “youth innovation” in STEM, applying classroom skills to a real green-tech challenge.
- **Next steps:** Interested users can explore the GitHub repository and project website for code, data, and further details (links provided on this slide). The model and documentation are open-source for collaboration and learning.